Week 4: Motor Control & H-Bridge - Theory Notes

EE0827 Embedded System Applications

Motor Control and H-Bridge Design

These notes complement the lecture slides with detailed calculations, component selection guidance, and practical implementation considerations for motor control systems.

Part 1

DC Motor Electrical Model

Equivalent Circuit

A DC motor can be modeled as a series combination of winding resistance, inductance, and back-EMF:

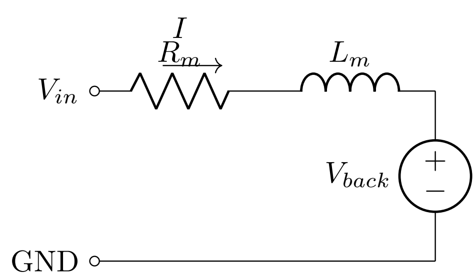


Figure 1: Motor Equivalent Circuit

$$V_{applied} = I \cdot R_m + L_m \frac{di}{dt} + V_{back}$$

At steady state (constant speed, $\frac{di}{dt} = 0$):

$$V_{applied} = I \cdot R_m + V_{back}$$

Motor Constants

Back-EMF constant ($\mathbf{K_v}$ or $\mathbf{K_e}$):

$$V_{back} = K_e \cdot \omega$$

Where ω is angular velocity in rad/s.

Torque constant ($\mathbf{K_t}$):

$$\tau = K_t \cdot I$$

Motor Constants Relationship

For a DC motor, $K_e = K_t$ (in SI units). This means the same constant describes both the voltage generated by rotation and the torque produced by current.

Calculating Motor Parameters

From datasheet or measurement:

1. **Winding resistance:** Measure with multimeter (motor stopped)
2. **Stall current:** $I_{stall} = V_{nom}/R_m$
3. **Back-EMF at no-load:** $V_{back} = V_{nom} - I_0 \cdot R_m$
4. **Back-EMF constant:** $K_e = V_{back}/\omega_{no-load}$

Example calculation:

Parameter	Value
Nominal voltage	6V
Winding resistance	7.5Ω
No-load current	70 mA
No-load speed	9000 RPM = 942 rad/s

$$V_{back} = 6V - 0.07A \cdot 7.5\Omega = 5.48V$$

$$K_e = \frac{5.48V}{942\,\mathrm{rad/s}} = 5.82\,\mathrm{mV/(rad/s)}$$

Current vs Load Relationship

As mechanical load increases:

1. Motor slows down → Back-EMF decreases
2. Lower back-EMF → Higher current flows
3. Higher current → More torque generated
4. Eventually reaches stall (zero speed, maximum current)

Key Insight: Current Indicates Load

Motor current is a direct indicator of mechanical load. This is the basis for software-based stall detection: monitor current and shut down if it exceeds a threshold for too long.

Part 2

Inductive Kickback & Flyback Protection

Energy Storage in Inductors

An inductor stores energy in its magnetic field:

$$E = \frac{1}{2}LI^2$$

Example: 10,mH motor inductance, 500,mA current:

$$E = \frac{1}{2} \cdot 0.01H \cdot (0.5A)^2 = 1.25\,\mathrm{mJ}$$

This energy must be dissipated when current is interrupted.

Voltage Spike Calculation

When the switch opens, inductor voltage:

$$V_L = -L\frac{di}{dt}$$

Worst case: Current drops to zero in switch opening time (~100,ns for MOSFET):

$$V_{spike} = L \cdot \frac{I}{t_{fall}}$$

Example: L = 10,mH, I = 500,mA, t_{fall} = 100,ns:

$$V_{spike} = 0.01H \cdot \frac{0.5A}{100 \cdot 10^{-9}s} = 50{,}000V$$

Flyback Voltage Can Destroy Components

The theoretical spike voltage can reach thousands of volts. In practice it is limited by component breakdown — but this is **destructive** breakdown (MOSFET avalanche, arc formation). Always install a flyback diode **before** applying power to inductive loads.

Flyback Diode Design

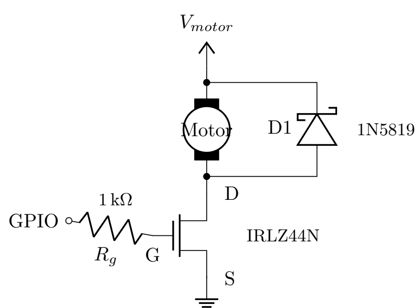


Figure 2: Low-Side Driver with Flyback Diode

Diode Selection Criteria

Parameter	Requirement	Reason
I_f(avg)	$> I_{motor}$ continuous	Normal freewheeling current
I_{fsm}	$> I_{stall}$	Peak current at motor stall
V_r	$> V_{motor}$	Reverse voltage during normal operation
t_{rr}	Fast	Important for high PWM frequency
V_f	Low	Reduces clamping voltage

Diode Types Comparison

Type	V_f	t_{rr}	Current	Best For
1N4001-4007	0.8-1.1V	30,µs	1A	Low-speed, simple
1N5817-5819	0.3-0.5V	$<$ 10,ns	1A	PWM motor control
SS12-SS14	0.3-0.5V	$<$ 10,ns	1A	SMD, PWM control
MBR150	0.5V	Fast	1A	Higher current margin

Why Schottky?

Schottky diodes have three advantages for motor flyback protection:

- **Lower forward voltage:** less power dissipation in the diode
- **No reverse recovery:** no shoot-through current spike during PWM switching
- **Faster clamping:** better protection during fast transients

Limitation: Lower reverse voltage ratings (typically 20–40 V).

Energy Dissipation in Flyback

Energy per switching cycle:

$$E_{cycle} = \frac{1}{2} L_m I^2$$

Power dissipated in diode:

$$P_{diode} = E_{cycle} \cdot f_{pwm} = \frac{1}{2} L_m I^2 \cdot f_{pwm}$$

Example: L_m = 5,mH, I = 300,mA, f = 5,kHz:

$$P_{diode} = \frac{1}{2} \cdot 0.005H \cdot (0.3A)^2 \cdot 5000Hz = 1.13W$$

Part 3

MOSFET Selection & Gate Drive

Key Selection Criteria

Parameter	Requirement	Reason
V_{ds}(max)	$> 2\times$ V_{motor}	Margin for transients
I_d(max)	$> 1.5\times$ I_{stall}	Startup and stall surges
V_{gs}(th)	< Logic voltage - 1V	Full turn-on at logic level
R_{ds}(on)	Low	Minimize power dissipation
Q_g	Low	Fast switching with weak driver

Logic-Level MOSFETs

Standard MOSFETs need 10V gate drive. For 3.3V MCU control, use logic-level MOSFETs:

Part Number	V_{ds}	I_d	V_{gs}(th)	R_{ds}(on)	Package
IRLZ44N	55V	47A	1-2V	22mΩ @ 5V	TO-220
IRLZ34N	55V	30A	1-2V	35mΩ @ 5V	TO-220
AO3400	30V	5.7A	0.65-1.5V	32mΩ @ 4.5V	SOT-23
Si2302	20V	2.6A	0.7-1.5V	60mΩ @ 4.5V	SOT-23

$R_{ds(on)}$ Depends on V_{gs}!

MOSFET on-resistance depends strongly on gate voltage. At 3.3 V gate drive, $R_{ds(on)}$ is significantly higher than the datasheet typical (usually specified at 10 V).

IRLZ44N typical values:

$V_{gs} = 2.7\,\text{V}$ $R_{ds(on)} = 75\,\text{m}\Omega$
$V_{gs} = 4.0\,\text{V}$ $R_{ds(on)} = 35\,\text{m}\Omega$
$V_{gs} = 5.0\,\text{V}$ $R_{ds(on)} = 22\,\text{m}\Omega$
$V_{gs} = 10\,\text{V}$ $R_{ds(on)} = 17.5\,\text{m}\Omega$

Power Dissipation Calculation

Conduction losses:

$$P_{cond} = I_d^2 \cdot R_{ds(on)}$$

Switching losses (during PWM):

$$P_{sw} = \frac{1}{2} V_{ds} \cdot I_d \cdot (t_{rise} + t_{fall}) \cdot f_{pwm}$$

Total dissipation:

$$P_{total} = P_{cond} + P_{sw}$$

Example: $\mathrm{I_d} = 500$,mA, $\mathrm{V_{ds}} = 12$V, $\mathrm{R_{ds(on)}} = 50$,mΩ, f = 10,kHz, $\mathrm{t_r} = \mathrm{t_f} = 50$,ns:

$$P_{cond} = (0.5A)^2 \cdot 0.05\Omega = 12.5\,\mathrm{mW}$$

$$P_{sw} = \frac{1}{2} \cdot 12V \cdot 0.5A \cdot 100ns \cdot 10kHz = 0.3\,\mathrm{mW}$$

$$P_{total} = 12.8\,\mathrm{mW}$$

(negligible)

Thermal Design

Junction temperature:

$$T_j = T_a + P_{total} \cdot R_{th(ja)}$$

Package	Typical $\mathrm{R_{th(ja)}}$
TO-220 (no heatsink)	62°C/W
TO-220 (with heatsink)	5-20°C/W
SOT-23	200-300°C/W
SOT-223	60-100°C/W

Example: 1W dissipation in TO-220, ambient 40°C:

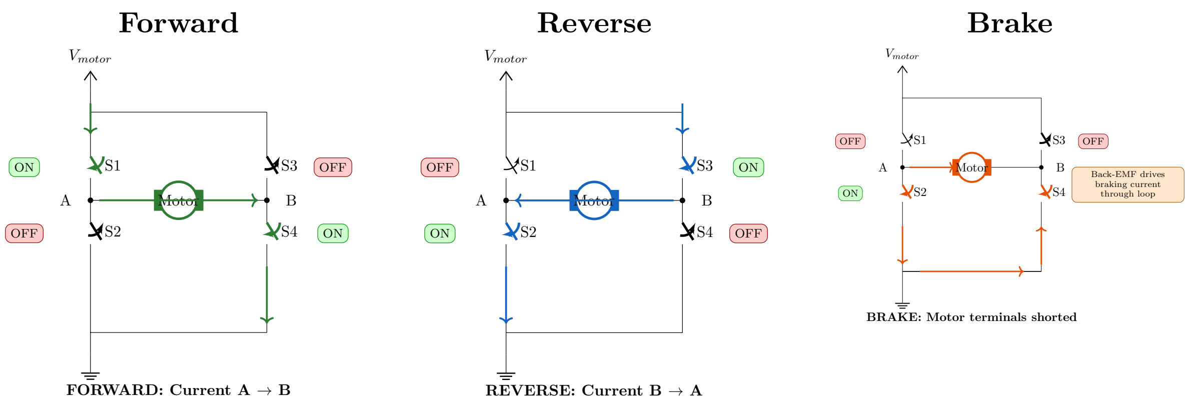
$$T_j = 40°C + 1W \cdot 62°C/W = 102°C$$

(within typical 150°C limit)

Part 4

H-Bridge Analysis

H-Bridge Operating Modes



Forward: $V_{motor} = V_{supply} - V_{ds(S1)} - V_{ds(S4)}$
Reverse: Current flows opposite direction through motor.
Brake: Both low-side ON — motor windings shorted, kinetic energy dissipated as heat.

Shoot-Through Current

If high-side and low-side switches on the same leg are both ON:

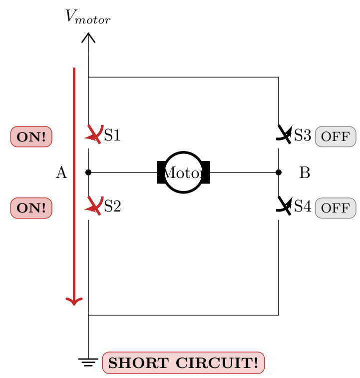


Figure 3: Shoot-Through Current Path

$$I_{shoot} = \frac{V_{supply}}{R_{ds(high)} + R_{ds(low)}}$$

Example: $V_{supply} = 12V$, R_{ds} total = 50,mΩ → $I_{shoot} = 240A$

Shoot-Through Destroys MOSFETs in Microseconds!

240 A through a small MOSFET causes instant thermal destruction. This is why H-bridge drivers include **dead-time** — a brief period where both switches are OFF during transitions.

Dead-Time Calculation

Required dead-time:

$$t_{dead} > t_{off(high)} + t_{on(low)} + t_{margin}$$

Where:

- t_{off}: Time for switch to fully turn off
- t_{on}: Time for switch to fully turn on
- margin: Safety factor (typically 50-100%)

Typical values: 0.5-2,µs for discrete MOSFETs, built into driver ICs.

Regenerative Braking

When motor spins and braking is applied, back-EMF becomes the source:

1. Motor generates voltage (back-EMF)
2. Current flows through body diodes (or synchronous rectification)
3. Energy returns to power supply

Capacitor sizing for regenerative current:

$$\Delta V = \frac{I_{regen} \cdot t_{brake}}{C_{supply}}$$

Large supply capacitors (100-1000,µF) prevent voltage spikes during regeneration.

Part 1

Motor Driver IC Comparison

L298N Voltage Drop

The L298N uses bipolar transistors (BJT) in the output stage:

$$V_{drop} = V_{ce(sat,high)} + V_{ce(sat,low)} \approx 1.4V + 1.4V = 2.8V$$

Actual motor voltage:

$$V_{motor} = V_{supply} - 2.8V$$

Example: 12V supply, 6V motor requires 8.8V supply minimum!

Power dissipation:

$$P_{L298N} = V_{drop} \cdot I_{motor} = 2.8V \cdot 1A = 2.8W$$

This is why L298N needs a heatsink.

TB6612FNG Analysis

MOSFET outputs have much lower drop:

$$V_{drop} = I \cdot (R_{ds(high)} + R_{ds(low)}) = I \cdot 1\Omega$$

At 1A: V_{drop} = 1V vs 2.8V for L298N.

Power dissipation:

$$P_{TB6612} = I^2 \cdot R_{ds} = (1A)^2 \cdot 1\Omega = 1W$$

TB6612FNG vs L298N: Nearly 3× More Efficient

The TB6612FNG dissipates 1 W vs 2.8 W for L298N at the same current. For battery-powered robots, this difference translates directly to longer run time.

Driver IC Selection Guide

Application	Recommended Driver	Reason
Battery robot	TB6612FNG	Low voltage drop, efficiency
High voltage (>13V)	L298N or DRV8871	Voltage rating
Very small motor	L293D or DRV8833	Appropriate current
High current (>2A)	DRV8871, BTS7960	Current rating
Precision control	TMC2209	Current regulation, quiet

Part 2

PWM Design Considerations

Frequency vs Inductance

Motor acts as a low-pass filter with corner frequency:

$$f_c = \frac{R_m}{2\pi L_m}$$

Example: R_m = 7.5,Ω, L_m = 5,mH:

$$f_c = \frac{7.5}{2\pi \cdot 0.005} = 239\,\text{Hz}$$

PWM frequency should be well above f_c for smooth current.

Current Ripple

Peak-to-peak current ripple:

$$\Delta I = \frac{V_{supply} \cdot D \cdot (1 - D)}{L_m \cdot f_{pwm}}$$

Example: V_{supply} = 12V, D = 50%, L_m = 5,mH, f = 5,kHz:

$$\Delta I = \frac{12V \cdot 0.5 \cdot 0.5}{0.005H \cdot 5000Hz} = 0.12A$$

(120,mA ripple)

Ripple Minimization

Current ripple is minimized by:

- Higher inductance (larger motor)
- Higher PWM frequency
- Duty cycle near 0% or 100%

PWM Resolution

Timer resolution for PWM:

$$Resolution = \frac{f_{timer}}{f_{pwm}}$$

Example: 84,MHz timer, 10,kHz PWM:

$$Resolution = \frac{84\text{MHz}}{10\text{kHz}} = 8400\,\text{steps}$$

Duty cycle resolution: 100%/8400 = 0.012% (excellent).

Minimum Duty Cycle

Motors have static friction (stiction) that must be overcome:

$$V_{min} = I_{starting} \cdot R_m$$

$$D_{min} = \frac{V_{min}}{V_{supply}}$$

Practical observation: Most motors need 15-25% duty cycle to start rotating.

Part 3

Current Sensing Design

Sense Resistor Selection

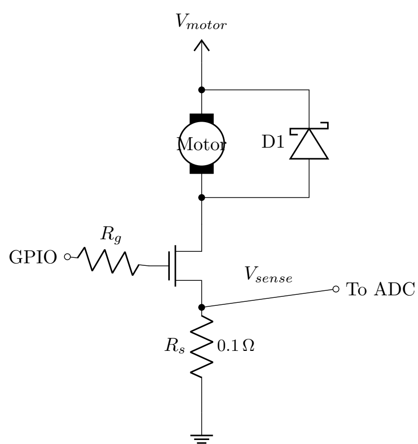


Figure 4: Motor Driver with Current Sensing

Parameter	Consideration
Resistance	Balance voltage drop vs signal level
Power rating	$P = I^2 \cdot R$
Inductance	Low inductance for fast response
Temperature coefficient	Stable over operating range

Typical values: 0.01,Ω to 0.5,Ω

Voltage Signal Calculation

At sense resistor:

$$V_{sense} = I_{motor} \cdot R_{sense}$$

ADC input (3.3V reference, 12-bit):

$$ADC_{value} = \frac{V_{sense}}{3.3V} \cdot 4096$$

$$I_{measured} = \frac{ADC_{value} \cdot 3.3V}{4096 \cdot R_{sense}}$$

Example: R_{sense} = 0.1,Ω, I_{motor} = 500,mA:

$$V_{sense} = 0.5A \cdot 0.1\Omega = 50\,\mathrm{mV}$$

$$ADC_{value} = \frac{0.05V}{3.3V} \cdot 4096 = 62\,\text{counts}$$

Low Resolution Without Amplification

62 counts out of 4096 gives very poor current measurement resolution. For practical current sensing, use a dedicated current sense amplifier (INA180, INA181, INA240) with 20–100· gain.

Current Sense Amplifier

Parameter	INA180A1	INA180A2	INA180A3
Gain	20 V/V	50 V/V	100 V/V
Bandwidth	160,kHz	68,kHz	35,kHz
Input offset	±150,µV	±150,µV	±150,µV

With INA180A2 (50 V/V):

$$V_{out} = 50 \cdot 0.05V = 2.5V$$

$$ADC_{value} = \frac{2.5V}{3.3V} \cdot 4096 = 3103\,\text{counts}$$

Much better resolution!

Overcurrent Detection

Software approach:

```
#define CURRENT_LIMIT_ADC  2500  // ~400mA with 0.1Ω, 50× gain
#define STALL_TIMEOUT_MS   100

void motor_protection_task(void) {
    uint16_t current_adc = read_motor_current();

    if (current_adc > CURRENT_LIMIT_ADC) {
        overcurrent_count++;
        if (overcurrent_count > STALL_TIMEOUT_MS) {
            motor_stop();
            fault_flag = FAULT_OVERCURRENT;
        }
    } else {
        overcurrent_count = 0;
    }
}
```

Hardware approach: Many driver ICs have a current limit pin or automatic shutdown.

Current Sense Resistor Values

Motor Current	R_{sense}	V_{sense} @ max
100,mA	1,Ω	100,mV
500,mA	0.2,Ω	100,mV
1,A	0.1,Ω	100,mV
2,A	0.05,Ω	100,mV
5,A	0.02,Ω	100,mV

Use amplifier (20-100× gain) for ADC with low R_{sense} values.

Part 4

Practical Implementation

Complete Low-Side Driver Circuit

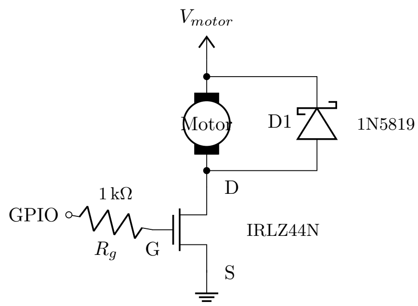


Figure 5: Low-Side MOSFET Motor Driver with Flyback

Component values:

- R_{gate}: 100,Ω - 1,kΩ (limits gate current, controls switching speed)
- D1: Schottky rated for $> I_{stall}$
- C1: 10-100,nF (high-frequency filtering, optional)
- R_{sense}: 0.05-0.5,Ω (for current measurement)

STM32 PWM Configuration

```
// Timer 2, Channel 1, 10 kHz PWM
void pwm_init(void) {
    RCC->APB1ENR |= RCC_APB1ENR_TIM2EN;

    TIM2->PSC = 0;                      // No prescaler
    TIM2->ARR = 8400 - 1;               // 84MHz / 8400 = 10 kHz
    TIM2->CCR1 = 0;                     // Start at 0% duty

    // PWM mode 1, preload enable
    TIM2->CCMR1 = TIM_CCMR1_OC1M_2 | TIM_CCMR1_OC1M_1 |
                  TIM_CCMR1_OC1PE;

    TIM2->CCER = TIM_CCER_CC1E;      // Enable output
```

```c
    TIM2->CR1 = TIM_CR1_CEN;         // Enable timer
}

void set_motor_speed(uint16_t duty_percent) {
    uint32_t ccr = (8400 * duty_percent) / 100;
    TIM2->CCR1 = ccr;
}
```

H-Bridge Control with TB6612FNG

TB6612FNG Pin	STM32 Pin	Function
PWMA	PA0 (TIM2_CH1)	Speed control
AIN1	PB4	Direction bit 1
AIN2	PB5	Direction bit 2
STBY	PA3	Standby (active low)
VM	External 6-12V	Motor power
VCC	3.3V	Logic power

```c
typedef enum {
    MOTOR_STOP, MOTOR_FORWARD, MOTOR_REVERSE, MOTOR_BRAKE
} motor_state_t;

void motor_set_state(motor_state_t state, uint16_t speed) {
    switch (state) {
        case MOTOR_STOP:
            GPIOA->BSRR = GPIO_BSRR_BR3;  // STBY low
            break;
        case MOTOR_FORWARD:
            GPIOA->BSRR = GPIO_BSRR_BS3;  // STBY high
            GPIOB->BSRR = GPIO_BSRR_BS4;  // AIN1 high
            GPIOB->BSRR = GPIO_BSRR_BR5;  // AIN2 low
            set_motor_speed(speed);
            break;
        case MOTOR_REVERSE:
            GPIOA->BSRR = GPIO_BSRR_BS3;  // STBY high
            GPIOB->BSRR = GPIO_BSRR_BR4;  // AIN1 low
            GPIOB->BSRR = GPIO_BSRR_BS5;  // AIN2 high
            set_motor_speed(speed);
            break;
```

```
        case MOTOR_BRAKE:
            GPIOA->BSRR = GPIO_BSRR_BS3;  // STBY high
            GPIOB->BSRR = GPIO_BSRR_BS4;  // AIN1 high
            GPIOB->BSRR = GPIO_BSRR_BS5;  // AIN2 high
            set_motor_speed(100);           // Full brake
            break;
    }
}
```

Troubleshooting Guide

Motor Not Running

Symptom	Possible Cause	Test	Solution
Motor silent	No power	Check V_{motor} with DMM	Verify supply
Motor silent	MOSFET not switching	Check gate voltage	Verify GPIO output
Motor silent	Open flyback diode	Check diode continuity	Replace diode
Motor hums	Stalled or blocked	Check if shaft rotates	Remove obstruction
Runs briefly	Overcurrent shutdown	Check current draw	Reduce load

MOSFET Getting Hot

Symptom	Possible Cause	Test	Solution
Hot during PWM	High switching losses	Measure with scope	Increase gate drive
Hot at low speed	Operating in linear region	Check V_{gs}	Use logic-level FET
Hot at full speed	High $R_{ds(on)}$	Calculate I^2R	Use lower R_{ds} FET
Very hot quickly	Shoot-through	Check H-bridge timing	Add dead-time

Motor Running Rough

Symptom	Possible Cause	Test	Solution
Jerky at low speed	PWM frequency too low	Listen for whine	Increase to >4,kHz
Speed fluctuates	Poor current regulation	Scope motor current	Add bulk capacitors
Audible whine	PWM in audible range	Change frequency	Use 15-20,kHz

Quick Reference

Formula Summary

Parameter	Formula
Stall current	$I_{stall} = V_{nom}/R_m$
Back-EMF	$V_{back} = V_{applied} - I \cdot R_m$
Inductor voltage	$V_L = -L \cdot di/dt$
Inductor energy	$E = \frac{1}{2}LI^2$
MOSFET power	$P = I^2 \cdot R_{ds(on)}$
PWM average voltage	$V_{avg} = D \cdot V_{max}$
Duty cycle	$D = t_{on}/T$
Current ripple	$\Delta I = V \cdot D \cdot (1 - D)/(L \cdot f)$

PWM Frequency Guidelines

Application	Frequency	Reason
DC motor speed	4-20,kHz	Inaudible, efficient
Servo PWM	50,Hz	Standard servo spec
LED dimming	100,Hz - 1,kHz	No flicker visible
Audio amplifier	$>$ 40,kHz	Above audible range